

VERSION 1.0 | APRIL 2025



Shilpa-Kala Showcase

Connecting Ancient Craft with the Modern World

Product Requirements Document

Android App Development Using Generative AI
Heritage Arts & Artisanal Commerce Domain

Prepared by: GenAI Android Development Team
Category: Heritage Arts · Artisanal Commerce · Cultural Preservation
Platform: Android (API 26+)

01 Executive Summary

Shilpa-Kala Showcase (meaning 'Art-Craft Exhibition' in Kannada/Sanskrit) is an AI-powered Android application designed to serve as a global digital gallery for India's traditional stone and wood carvers — particularly those in heritage clusters such as Shivarapatna, Belur, and Halebidu in Karnataka. The app bridges a critical gap in artisanal commerce by providing master craftsmen (Shilpis) a verified digital presence, enabling national and international buyers to discover, appreciate, and commission authentic traditional sculptures.

Attribute	Details
Product Name	Shilpa-Kala Showcase — Digital Gallery for Master Craftsmen
Domain	Heritage Arts · Artisanal Commerce · Cultural Preservation
Platform	Android (API Level 26 / Android 8.0 and above)
Primary Users	Traditional stone/wood carvers (Shilpis), art buyers (national & international)
Secondary Users	Art collectors, cultural NGOs, tourism departments, export agents
Language Support	English, Kannada (Phase 1); Hindi + 4 regional languages (Phase 2)
GenAI Layer	AI-generated heritage narratives, carving style descriptions, buyer query responses
Document Version	1.0 — April 2025

02 Problem Statement

India is home to thousands of master craftsmen who inherit centuries-old techniques — Hoysala stone carving, Channapatna wood turning, Mysore sandalwood inlay — that rival the finest art in the world. Yet these artisans remain invisible to the market that would most value their work. The digital divide, lack of marketing knowledge, and absence of a trusted platform leave master Shilpis entirely dependent on middlemen who capture most of the value of their labour.

No Digital Presence

Stone carvers in places like Shivarapatna have no website, no social media profile, and no way for a buyer in Mumbai, London, or New York to find them. Word-of-mouth within the village is the only marketing channel available.

Middlemen and Value Leakage

Without direct buyer access, Shilpis sell to dealers at a fraction of market price. A sculpture worth Rs. 50,000 in a Bangalore gallery may earn the carver only Rs. 5,000 — a 90% value loss with no recourse.

Art is Dying with the Artisans

The younger generation sees no economic future in traditional carving. With no 'modern' way to market their skills and no documented proof of their artistry, the craft faces generational extinction. Entire carving techniques — some dating to the 12th century Hoysala Empire — risk being lost forever.

Buyers Lack Discovery and Trust

National and international buyers interested in authentic Indian stone sculpture have no verified, centralised platform to discover artisans, understand the craft's heritage, or initiate a commission inquiry with confidence.

Craft Heritage is Undocumented

The visual story of how a block of raw stone becomes a masterpiece — the weeks of chiselling, the precision of each cut — is never documented. This 'process story' is itself a piece of cultural heritage and a powerful sales narrative that is currently lost.

03 Vision & Objectives

Vision: Every master Shilpi in India, regardless of their village's remoteness or their digital literacy, deserves a global window for their art — a platform as refined as the sculptures they create.

Strategic Objectives

- Empower artisans by providing a premium digital gallery that gives them a professional identity and direct access to buyers — eliminating middlemen.
- Preserve cultural heritage by documenting the process, history, and techniques of traditional carving styles (Hoysala, Dravidian, Vijayanagara) through GenAI-curated narratives.
- Facilitate 'Make in India' commerce by enabling international export inquiries through a streamlined, WhatsApp-integrated contact flow.
- Bridge the digital divide by providing an ultra-simple upload and profile interface operable by artisans with minimal smartphone literacy.
- Create a trusted marketplace by verifying Shilpi identity and providing transparent product IDs for each artwork — enabling traceable, authentic transactions.

04 Scope of the Application

4.1 In-Scope (Phase 1 — MVP)

- Shilpi (artisan) profile: name, village, specialisation, years of experience, and profile photo.
- Artwork portfolio gallery: high-resolution image upload with title, material, dimensions, and price range.
- Work-in-Progress (WIP) timeline: multi-photo documentation of a sculpture from raw stone to finished art.
- Order Inquiry: pre-formatted WhatsApp message trigger with artwork Product ID embedded.

- Heritage Story section: GenAI-generated cultural narratives for each carving style.
- Zoom & pan on artwork images for detailed carving inspection.
- English and Kannada language support.
- Offline browsing of cached portfolio content.

4.2 Out-of-Scope (Phase 1)

- In-app payment gateway or direct e-commerce transactions.
- Live video or AR sculpture visualisation.
- Multi-Shilpi management (artist co-operative dashboard) — Phase 2.
- Integration with government export portals (DGFT, APEDA) — Phase 2.
- iOS application.
- Auction or bidding mechanism.
- Buyer account creation or wishlists.

4.3 Future Scope (Phase 2 onwards)

- AI-powered sculpture authentication and provenance certificate generation.
- AR viewer — place a virtual sculpture in the buyer's room using the phone camera.
- Artist co-operative dashboard for NGOs to manage 20+ Shilpi profiles.
- Export inquiry routing to MSME Udyam portal.
- Regional language packs: Tamil, Telugu, Hindi, Odia, Marathi.
- Craft video documentation — short-form process videos with GenAI narration.

05 User Personas

Attribute	Ramu, 52 — Shilpi	Priya, 34 — Urban Buyer	Kavitha, 41 — NGO Coordinator
Context	Stone carver in Shivarapatna. 30 yrs experience. Basic Android. No internet marketing knowledge.	Interior designer in Bengaluru. Looks for authentic art for clients. Active on Instagram.	Runs a craft promotion NGO in Mysuru. Manages 15 artisan families. Needs reporting tools.
Goals	Upload photos of work. Get buyer inquiries. Show WIP process.	Discover verified artisans. Zoom into carving detail. Inquire directly with product reference.	Create profiles for artisans. Track inquiries. Share portfolio links.
Frustrations	Complex upload flows. English-only apps. No receipt confirmation.	Fake or low-quality craft listings. No provenance info. No direct contact.	No bulk profile creation. No inquiry analytics.

06 Functional Requirements

FR-01: Shilpi Profile & Onboarding

ID	Priority	Requirement
FR-01.1	Must	App shall allow creation of a Shilpi profile with: full name, village/district, carving specialisation (stone/wood/ivory), years of experience, and profile photo.
FR-01.2	Must	Profile shall be persisted locally in Room DB and survive app restart without requiring re-entry.
FR-01.3	Must	Onboarding flow shall be completable in under 3 minutes with a 4-step guided wizard.
FR-01.4	Must	Each Shilpi profile shall receive a unique, human-readable Shilpi ID (e.g., SK-KA-0042) for buyer reference.
FR-01.5	Should	Shilpi shall be able to add a 30-second audio introduction in Kannada or English — played on profile view.
FR-01.6	Could	NGO co-ordinator (admin) mode shall allow bulk creation of up to 20 Shilpi profiles (Phase 2).

FR-02: Artwork Portfolio Gallery

ID	Priority	Requirement
FR-02.1	Must	Shilpi shall be able to upload up to 30 high-resolution images per artwork listing, with title, material, approximate dimensions (H x W x D in cm), and indicative price range.
FR-02.2	Must	Each artwork shall be assigned a unique Product ID (e.g., SKP-KA-0042-007) displayed on its detail screen.
FR-02.3	Must	Gallery shall render using Glide with progressive loading, handling 20+ images without UI lag or OOM crash.
FR-02.4	Must	Image viewer shall support pinch-to-zoom (min 1x, max 5x) and double-tap zoom for carving detail inspection.
FR-02.5	Must	Gallery shall display artworks in a 2-column masonry grid on the Shilpi portfolio screen.
FR-02.6	Should	Buyer shall be able to filter artworks by material (Stone / Wood / Mixed), style (Hoysala / Dravidian / Folk / Contemporary), and approximate price range.
FR-02.7	Should	Shilpi shall be able to mark an artwork as 'Sold,' 'Available,' or 'Commission-Only.'
FR-02.8	Could	Artwork listings shall be shareable as a deep-link URL or image card via WhatsApp and other share targets.

FR-03: Work-in-Progress (WIP) Timeline

ID	Priority	Requirement
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FR-03.1	Must	Shilpi shall be able to create a WIP project linked to an artwork, adding dated photos at each stage (Raw Stone → Rough Shaping → Detail Carving → Finishing → Completed).
FR-03.2	Must	WIP timeline shall be displayed as a horizontal scrollable photo strip with stage labels and upload dates.
FR-03.3	Must	Each WIP stage photo shall support a short caption (max 100 characters).
FR-03.4	Should	GenAI shall auto-generate a descriptive narrative for the WIP journey ('This Nandi bull began as a 200 kg Deccan granite block...') based on stage photos and Shilpi inputs.
FR-03.5	Could	Completed WIP timeline shall be exportable as a shareable image collage — suitable for WhatsApp or Instagram.

FR-04: Order Inquiry (WhatsApp Integration)

ID	Priority	Requirement
FR-04.1	Must	Each artwork detail screen shall display a prominent 'Enquire' button that launches WhatsApp with a pre-formatted inquiry message.
FR-04.2	Must	Pre-formatted message shall include: Artwork Product ID, artwork title, Shilpi name, and buyer's interest statement. Example: 'Hello, I am interested in [Product ID: SKP-KA-0042-007] — Nandi Bull in Granite by Ramu Shilpi. Please share availability and pricing.'
FR-04.3	Must	App shall use Android Intent to launch WhatsApp. If WhatsApp is not installed, a fallback SMS/email intent shall be triggered.
FR-04.4	Should	Inquiry button tap shall log an inquiry event locally (timestamp, product ID) for Shilpi to view as an 'Interest Log.'
FR-04.5	Could	App shall support a 'Request Custom Commission' flow — a multi-field form (dimensions, material, theme, budget) generating a formatted inquiry message.

FR-05: Heritage Story (GenAI-Powered)

ID	Priority	Requirement
FR-05.1	Must	Each carving style (Hoysala, Dravidian, Vijayanagara, Channapatna, Folk) shall have a Heritage Story card with GenAI-generated narrative covering historical origins, technique description, and cultural significance.
FR-05.2	Must	Heritage Story shall be accessible from the artwork detail screen ('About this Style' button) and from a dedicated 'Heritage' tab.
FR-05.3	Must	All GenAI content shall be clearly labelled 'AI-generated heritage content — verified by cultural experts.'

FR-05.4	Should	GenAI shall answer free-text buyer queries about a sculpture ('What is the significance of the Nandi bull in Shaivite tradition?') with contextually appropriate responses.
FR-05.5	Should	Common heritage Q&As shall be pre-cached offline — at least 30 entries covering major carving styles.
FR-05.6	Could	GenAI shall generate a personalised 'Collector's Certificate' text for a purchased artwork — suitable for framing alongside the sculpture.

FR-06: Discovery & Search

ID	Priority	Requirement
FR-06.1	Must	App shall provide a searchable directory of Shilpi profiles filterable by district/region, carving material, and style.
FR-06.2	Must	Search shall return results within 1 second for a local Room DB query across up to 200 Shilpi profiles.
FR-06.3	Should	App shall feature a 'Featured Artisan' section on the home screen, rotating weekly to highlight a different Shilpi and their signature work.
FR-06.4	Should	Buyer shall be able to save artworks to a local 'Wishlist' for offline reference.

FR-07: Settings & Accessibility

ID	Priority	Requirement
FR-07.1	Must	Language selection between English and Kannada shall be available on first launch and in Settings.
FR-07.2	Must	All critical text shall meet WCAG AA contrast standards. Gallery background shall be neutral (off-white or charcoal) to make sculpture colours stand out.
FR-07.3	Should	Font size shall be adjustable (Standard / Large / Extra-Large) for older Shilpi users.
FR-07.4	Should	Full offline mode — all cached portfolio content and Heritage Stories shall be browsable without internet.
FR-07.5	Could	Shilpi profile shall be shareable as a digital 'Business Card' — QR code that opens their app profile.

07 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Gallery of 20 high-resolution images (avg 3 MB each) shall load and display within 3 seconds on a mid-range device (3 GB RAM, Android 8.0).

NFR-02	Performance	Pinch-to-zoom interaction shall respond within 16ms (60fps) for smooth inspection of carving detail.
NFR-03	Performance	App cold start time ≤ 3 seconds on mid-range device.
NFR-04	Offline	All cached Shilpi profiles, artwork galleries, and Heritage Stories shall be fully accessible without network after initial sync.
NFR-05	Storage	App APK size ≤ 25 MB. Locally cached images shall use Glide's disk cache with a 200 MB cap.
NFR-06	Security	Shilpi personal data (phone number, address) stored locally in Room DB with SQLCipher encryption. Not transmitted without explicit consent.
NFR-07	Privacy	App shall comply with India's Digital Personal Data Protection Act (DPDPA) 2023. WhatsApp inquiry uses device's installed app — no contact data stored server-side.
NFR-08	Usability	Shilpi upload flow (photo + details for one artwork) shall be completable in under 4 minutes by a first-time user with no tutorial.
NFR-09	Accessibility	Minimum font size 14sp. All interactive elements ≥ 48 dp touch target. TalkBack screen reader support for artwork descriptions.
NFR-10	Reliability	App crash-free rate $\geq 99.5\%$ across Xiaomi MIUI, Samsung One UI, and Realme ColorOS — the top 3 Indian OEM skins.
NFR-11	Scalability	Architecture shall support addition of 5 regional languages without code restructure — string resource-driven i18n from day one.
NFR-12	Maintainability	Code coverage $\geq 70\%$ via unit + integration tests. All modules MVVM-compliant with Hilt dependency injection.
NFR-13	Compatibility	Minimum SDK: API 26 (Android 8.0). Target SDK: API 34 (Android 14).

08 Feature Catalogue

8.1 Must-Have Features (MVP)

Feature	Description
Shilpi Profile	Verified artisan profile with Shilpi ID, specialisation, region, experience, and profile photo. Foundation of trust and discoverability.
Portfolio Gallery	Masonry grid of high-res artwork images with Glide-powered loading. Handles 20+ images smoothly. Pinch-to-zoom for detail inspection.
Artwork Product ID	Auto-generated unique ID per artwork — embedded in inquiry messages for traceable transactions.
WIP Timeline	Multi-stage photo documentation of a sculpture's journey from raw material to finished work. Horizontal scrollable strip.
WhatsApp Inquiry	One-tap 'Enquire' button launching WhatsApp with pre-formatted message including Product ID. Fallback to SMS/email.

Heritage Story	GenAI-curated cultural narratives for each carving tradition — history, technique, significance. Pre-cached offline.
Discovery & Search	Filter Shilpis by region, material, and style. Returns results in under 1 second from local Room DB.
English + Kannada	Full UI and content in English and Kannada with in-app language switch — accessible to Shilpis and local buyers.
Offline Mode	All cached portfolio content browsable without internet after initial load.

8.2 Good-to-Have Features (Phase 1 Stretch / Phase 2)

Feature	Description
GenAI Chatbot	Free-text Q&A on heritage, technique, and sculpture significance. Powered by Gemini API — falls back to cached Q&A offline.
Custom Commission Flow	Structured inquiry form (dimensions, material, theme, budget) generating a WhatsApp commission brief with all specs.
Collector's Certificate	GenAI-generated provenance and collector's narrative for a purchased artwork — shareable as PDF or image.
WIP Shareable Collage	Auto-generated image collage of WIP timeline stages — shareable to Instagram/WhatsApp for artisan marketing.
Wishlist	Local bookmark of artworks for buyers to save and revisit offline without account creation.
Shilpi QR Card	QR code that deep-links to the Shilpi's in-app profile — printable as a business card for exhibitions.
AR Sculpture Viewer	Place a virtual 3D model of a sculpture in the buyer's space using ARCore (Phase 2).
Export Portal Integration	Route international buyer inquiries to MSME Udyam/APEDA for formalised export (Phase 2).
Multi-language Pack	Tamil, Telugu, Hindi, Odia, Marathi — via downloadable string resource packs (Phase 2).

09 App Usage & User Flow

Shilpi Onboarding & Upload Flow

- 1. Shilpi opens app → selects 'I am an Artisan' → 4-step wizard: Name → Village & District → Specialisation → Profile Photo.
- 2. Shilpi ID (e.g., SK-KA-0042) auto-generated and displayed — 'Your Artist ID.'
- 3. Shilpi taps 'Add Artwork' → uploads photos → fills title, material, dimensions, price range → submits.
- 4. Product ID auto-generated (e.g., SKP-KA-0042-007) — displayed on artwork card.

- 5. Shilpi taps 'Add WIP Timeline' → uploads staged photos → adds captions for each stage.

Buyer Discovery & Inquiry Flow

- 1. Buyer opens app → sees 'Featured Artisan' on Home + search bar.
- 2. Buyer searches 'Hoysala stone carving' → filtered Shilpi list displayed.
- 3. Buyer taps Shilpi card → views portfolio gallery in masonry grid.
- 4. Buyer taps artwork → full-screen viewer. Pinch to zoom into carving detail.
- 5. Buyer taps 'Enquire' → WhatsApp launches with pre-filled message including Product ID.
- 6. Buyer taps 'About this Style' → Heritage Story card with GenAI narrative appears.

WIP Timeline Flow

- 1. Buyer views artwork detail → taps 'See how it was made.'
- 2. Horizontal scrollable WIP strip loads: Day 1 (raw granite block) → Day 15 (rough shape) → Day 45 (detailed carving) → Day 60 (finished Nandi bull).
- 3. GenAI-generated narrative auto-plays as a caption below: 'This 200 kg Deccan granite block was sourced from Ramanagara quarry. Ramu Shilpi used a 3mm chisel to...'

Heritage Story & GenAI Q&A Flow

- 1. Buyer taps 'Heritage' tab → list of carving traditions (Hoysala, Dravidian, Channapatna, etc.).
- 2. Buyer taps 'Hoysala Style' → full-screen Heritage Story card with origins (12th century), technique, famous temples (Belur, Halebidu), and cultural significance.
- 3. Buyer types question: 'What stone is used in Hoysala carvings?' → GenAI responds: 'Hoysala sculptors used chloritic schite (soapstone)...' within 3 seconds.
- 4. Response footer: 'AI-generated heritage content — verified by cultural experts.'

10 Technical Implementation

10.1 Architecture

The app shall follow the MVVM (Model-View-ViewModel) pattern using Android Jetpack components. This ensures clean separation of concerns, testability, and lifecycle awareness — critical for managing large image sets and asynchronous GenAI API calls.

Layer	Technology	Responsibility
UI Layer	Jetpack Compose / XML Layouts	Artwork gallery, profile screens, WIP timeline — premium art gallery aesthetic
ViewModel Layer	Android ViewModel + StateFlow	Business logic, UI state, GenAI call orchestration, search filtering

Repository Layer	Repository Pattern	Single source of truth — Room DB (local) + optional Gemini API (remote)
Data Layer (Local)	Room DB + DataStore	Shilpi profiles, artwork listings, WIP entries, wishlist, GenAI cache
Image Loading	Glide	Progressive loading of high-res sculpture photos with disk cache (200 MB cap)
GenAI Layer	Gemini API / Gemini Nano	Heritage narratives, buyer Q&A, WIP story generation, commission text
WhatsApp	Android Intent (ACTION_VIEW)	Pre-formatted inquiry message with Product ID — zero in-app payment flow
Search	Room DB FTS4	Full-text search across Shilpi names, village, style, material

10.2 Key Library Decisions

Library	Purpose & Rationale
Glide	Industry-standard image loading for Android. Handles progressive JPEG loading, disk caching, and memory management for 20+ high-res sculpture images without OOM crashes.
MPAndroidChart (optional)	If analytics screen is added — inquiry trends, view counts per artwork for Shilpi dashboard.
Room DB	Local SQLite abstraction for offline-first Shilpi profiles and artwork data. DPDPA-compliant — no data leaves device without consent.
Retrofit + OkHttp	Remote API calls to Gemini for heritage narrative generation and buyer Q&A — used only when internet available.
Hilt (DI)	Dependency injection for testability, modular architecture, and clean ViewModel-Repository separation.
PhotoView	Advanced pinch-to-zoom and pan gesture handling for full-screen sculpture image inspection. Supports 1x–5x zoom.
Gemini API (Google)	On-demand generation of Heritage Story narratives, WIP journey descriptions, and collector certificates.
iText (PDF)	Artwork portfolio export and Collector's Certificate generation — shareable via WhatsApp or email.

10.3 Data Model (Key Entities)

Entity	Key Attributes
ShilpiProfile	id, shilpild (SK-KA-XXXX), name, village, district, state, specialisation, yearsExperience, bio, phoneNumber (encrypted), profilePhotoUri, audioIntroUri, createdAt

Artwork	id, productId (SKP-KA-XXXX-YYY), shilpId, title, material, style, heightCm, widthCm, depthCm, weightKg, priceRangeMin, priceRangeMax, status [AVAILABLE/SOLD/COMMISSION_ONLY], description, createdAt
ArtworkImage	id, artworkId, imageUri, sortOrder, caption, uploadedAt
WipProject	id, artworkId, title, genAiNarrative, createdAt
WipStage	id, wipProjectId, stageName [RAW/ROUGH/DETAIL/FINISHING/COMPLETE], photoUri, caption, stageDate
HeritageTopic	id, styleKey, styleName, originEra, regionOfOrigin, techniqueDescription, culturalSignificance, famousExamples, genAiContent, isCached, lastUpdated
InquiryLog	id, artworkId, productId, timestamp, inquiryType [WHATSAPP/SMS/EMAIL]
GenAIConversation	id, topicKey, userQuery, aiResponse, language, timestamp, isCached

11 UI / UX Guidelines

11.1 Design Philosophy

The UI must feel like a premium art gallery — minimalist, clean, and sophisticated — not a marketplace bazaar. Every screen must let the sculpture be the hero. The background recedes; the art speaks. Simultaneously, the Shilpi upload interface must be the opposite of intimidating — as simple as taking a photo and adding a label.

Core Design Principles

- Gallery Palette: Off-white (#F5F0EB) and deep charcoal (#1A1A2E) backgrounds to showcase sculpture tones. Gold (#B8860B) as the accent colour — evoking temple gold and craftsmanship.
- Sculpture-First Layout: Full-bleed images with minimal chrome. Title and price below, not overlaid. Zero visual noise around the artwork.
- Large Touch Targets: All buttons $\geq 48\text{dp}$. Primary CTA (Enquire) $\geq 60\text{dp}$ height. Critical for older Shilpi users.
- Typography Contrast: Serif display font for artwork titles (premium feel). Sans-serif for functional UI (readability). Strict WCAG AA compliance.
- Tactile Feedback: Every tap gets a visual response — subtle ripple on cards, scale animation on image selection, colour pulse on 'Enquire.'
- Zero Clutter: One primary action per screen. Advanced options behind 'More' or secondary buttons. Shilpi upload wizard shows one field at a time.
- Safe Colour Coding: Gold border = Featured/Verified. Green badge = Available. Grey badge = Sold. Amber badge = Commission Only.

11.2 Screen Inventory

Screen	Key Elements
Splash / Onboarding	App logo (chisel-and-stone motif), tagline 'Ancient Craft. Global Stage.', language selection, 3-screen onboarding carousel with illustration
Shilpi Onboarding Wizard	4-step form: Name → Village → Specialisation → Photo. Progress bar visible. Simple icon-led UI — operable by low-digital-literacy users
Home Screen	Featured Artisan card, search bar, category tiles (by style/material), recently added artworks strip
Shilpi Profile	Cover photo, Shilpi ID badge, specialisation tags, experience chip, portfolio gallery grid below, 'Contact Artisan' CTA
Artwork Detail	Full-screen hero image, zoom button, product ID chip, WIP button, Heritage Story button, 'Enquire on WhatsApp' primary CTA
Portfolio Gallery	2-column masonry grid, Glide-loaded images, status badge (Available / Sold / Commission), tap to artwork detail
WIP Timeline	Horizontal scrollable photo strip, stage labels, GenAI narrative panel below, share button
Heritage Story	Dark-background immersive card, era badge, regional map illustration, GenAI narrative paragraphs, Q&A input at bottom
Search & Discovery	Filter chips (style, material, region), Shilpi card list, result count
Settings	Language, font size, cache management, about/version, data export

12 Impact Goals

Impact Area	Description
Artisanal Empowerment	Shilpis receive a global digital window for their skills — eliminating middlemen and enabling direct buyer contact. A sculptor earning Rs. 5,000 via dealer can potentially earn Rs. 45,000 by selling directly, transforming livelihoods.
Heritage Preservation	The WIP Timeline feature documents rare carving techniques — some unchanged since the 12th century Hoysala Empire — creating a living digital archive of intangible cultural heritage for future generations.
Make in India & Export	Facilitating international buyer discovery and inquiry routing enables high-value craft exports. Indian stone sculptures command premium prices in international art markets — a largely untapped 'Make in India' export opportunity.
Generational Continuity	A digital-first platform showing traditional carving as a viable, globally connected profession gives the younger generation a reason to learn the craft — combating generational skill extinction.
Rural Economic Development	Artisan clusters like Shivarapatna, Belur, and Channapatna receive indirect economic uplift as buyer interest, commission volumes, and craft tourism increase through digital visibility.

13 Success Criteria

13.1 Academic / Student Deliverable Criteria

Criterion	Measurable Benchmark
Image Performance	Gallery of 20 high-res images loads without lag or OOM crash on a 3 GB RAM device. Verified via Android Profiler memory trace.
Product ID in Inquiry	The 'Enquire' button pre-fills WhatsApp message with the correct Product ID of the viewed artwork — validated for 10 different artworks.
Pinch-to-Zoom	Artwork image viewer supports pinch-to-zoom from 1x to 5x with smooth 60fps gesture response — no frame drops visible.
WIP Timeline	Shilpi can create a WIP project with at least 5 staged photos and captions. Timeline renders correctly with stage labels.
Heritage Story (GenAI)	GenAI assistant generates a contextually accurate Heritage Story for at least 3 distinct carving styles in a live demo.
WhatsApp Intent	Enquire button correctly launches WhatsApp (or SMS fallback) on a device with WhatsApp installed and one without.
UI Quality	UI passes peer review for premium gallery aesthetic: minimalist, clean, sculpture-forward. No cluttered layouts or garish colours.
Offline Browsing	Cached Shilpi profiles and artworks are fully browsable with Airplane Mode enabled after initial load.
Code Quality	MVVM architecture, Room DB, Glide, and Hilt used as specified. Unit tests cover $\geq 70\%$ of ViewModel layer.

13.2 Product Success Metrics (Post-Launch)

Metric	Target
D30 Retention (Buyer)	$\geq 35\%$ of buyers who browsed a Shilpi portfolio return within 30 days
Inquiry Conversion Rate	$\geq 15\%$ of artwork detail screen views result in a WhatsApp inquiry tap
Shilpi Profile Completeness	$\geq 80\%$ of registered Shilpis have at least 5 artwork listings and 1 WIP project
Image Load Success Rate	$\geq 99\%$ of gallery image loads complete without error or placeholder fallback (cached content)
GenAI Query Satisfaction	$\geq 75\%$ positive feedback (thumbs up) on GenAI Heritage Story and Q&A responses
Crash-Free Rate	$\geq 99.5\%$ crash-free sessions on Play Store
Offline Usage	$\geq 60\%$ of buyer browse sessions occur with no network connection

14 Risks & Mitigations

Risk	Severity	Description	Mitigation
GenAI Heritage Accuracy	HIGH	GenAI may generate historically inaccurate content about carving traditions, damaging credibility.	All heritage narratives reviewed by a cultural expert before caching. Label all GenAI content. Fallback to pre-verified offline cards.
Low Shilpi Digital Literacy	HIGH	Artisans unable to operate upload interface may abandon onboarding, leaving the platform empty.	Icon-first wizard with one field per screen. Optional NGO/co-ordinator assisted onboarding. Field testing with 5 real Shilpis before launch.
Image Quality & Size	MEDIUM	Low-res or oversized images from artisan phones may degrade gallery quality or cause OOM crashes.	Glide auto-downsample on load. Upload-time image compression (max 2 MB per image). Guidelines shown during upload: 'Best angle, good light.'
WhatsApp Not Installed	MEDIUM	Significant fraction of devices may not have WhatsApp — breaking the primary inquiry flow.	Implement Intent chooser fallback: WhatsApp → SMS → Email. Test all three paths on devices without WhatsApp.
Device Fragmentation	MEDIUM	Glide or zoom library may behave differently on MIUI, One UI, or ColorOS camera/storage permissions.	Test on Xiaomi Redmi (MIUI), Samsung Galaxy A-series (One UI), Realme (ColorOS). Fix OEM-specific permission dialogs.
Data Privacy (Shilpi Phone No.)	HIGH	Shilpi phone number used for WhatsApp is sensitive PII. Exposure risk if device is lost or app is decompiled.	Encrypt phone number in Room DB with SQLCipher. Never display full number on screen — mask as '98xxxx789.' Only used for WhatsApp Intent.
GenAI API Cost	MEDIUM	Gemini API calls for heritage generation at scale may incur unexpected costs.	Cache all generated heritage content permanently after first call. Use Gemini Nano on-device for common Q&As. Remote API only for novel queries.

15 Appendix

A. Karnataka Traditional Carving Styles — Reference Guide

Style	Era	Key Regions	Primary Material & Notes
Hoysala	12th–14th century CE	Belur, Halebidu, Somnathapura (Karnataka)	Chloritic schite (soapstone) — softens on quarrying, hardens on exposure. Renowned for intricate frieze work.
Dravidian (Chola)	7th–13th century CE	Tamil Nadu, South Karnataka	Granite — extremely hard. Bronze casting also prominent. Bold, large-form sculptures of deities.
Vijayanagara	14th–16th century CE	Hampi, Lepakshi (Karnataka/AP)	Granite. Monolithic carvings. Massive scale — Ugra Narasimha at Hampi is 6.7m tall.
Channapatna	16th century onwards	Channapatna, Ramanagara (Karnataka)	Ivory wood (aale mara). Turned on lathe. Bright lacquer finish. Mainly toys and decorative items.
Mysore Sandalwood	Royal era, 16th c+	Mysore, Bangalore (Karnataka)	Pure sandalwood — highly regulated. Intricate inlay and low-relief carving. Export-controlled.
Folk / Tribal	Pre-historic–ongoing	Dakshina Kannada, Kodagu, Uttara Kannada	Local stone, terracotta, wood. Votive figurines, hero stones, deity idols. Raw, expressive aesthetic.

B. Glossary

Term	Definition
Shilpi	Sanskrit/Kannada term for a sculptor or craftsman. Used throughout to refer to the artisan user.
Shilpa-Kala	Sanskrit compound: Shilpa (craft/art) + Kala (art/skill). Literally 'the art of craftsmanship.'
Hoysala Style	Architectural and sculptural tradition of the Hoysala Empire (Karnataka, 12th–14th c.) — noted for densely detailed friezes on soap stone.
Chloritic Schite	The scientific name for the soapstone used in Hoysala carvings. Soft when quarried, hardens after prolonged exposure to air.
WIP Timeline	Work-in-Progress Timeline — multi-stage photo documentation of a sculpture from raw material to finished artwork.
Product ID	Unique identifier assigned to each artwork listing (e.g., SKP-KA-0042-007) for traceable inquiry and provenance.
Shilpi ID	Unique identifier assigned to each verified artisan profile (e.g., SK-KA-0042).
MVVM	Model-View-ViewModel — Android architectural pattern enforcing separation of UI, business logic, and data.
Room DB	Android Jetpack's SQLite ORM for structured local data storage — used for offline-first artisan data.

Glide	Open-source Android image loading library supporting progressive JPEG loading, caching, and memory management.
DPDPA	Digital Personal Data Protection Act, 2023 — India's data privacy law governing collection and storage of personal data.
GenAI	Generative AI — AI system (Gemini) powering heritage narrative generation and buyer Q&A chatbot.
WhatsApp Intent	Android Intent using ACTION_VIEW with a WhatsApp URI scheme to launch WhatsApp with pre-filled text — no API key or account required.
Make in India	Government of India initiative to encourage domestic manufacturing and export of Indian products — including traditional crafts.

Document Control

- Version 1.0 — Initial PRD Draft — April 2025
- Prepared for: GenAI Android App Development Course — Heritage Arts Track
- Review cadence: Every sprint (2 weeks) during development phase
- Next scheduled review: May 2025 — post-prototype usability testing with Shilpi community